Chilli fungal diseases

1. Damping off - *Pythium aphanidermatum*, *P. indicum*, , *Phytophthora parasitica*, *Rhizoctonia solani* and *Sclerotium rolfsii*.
2. Die back and fruit rot - *Colletotrichum capsici* (Tel: *Glomerella cingulata)*
3. Cercospora leaf spot/ frog eye leaf spot - *Cercospora capsici*
4. Alternaria leaf spot - *Alternaria solani*
5. Sclerotial wilt or Stem Rot - *Sclerotium rolfsii*
6. Verticillium wilt - *Verticillium dahliae, V. albo-atrum*
7. Fusarium wilt -*Fusarium oxysporum* f. sp. *capsici*
8. Powdery mildew - *Leveillula taurica* (anamorph - *Oidiopsis taurica*)

Chilli diseases

Bacterial diseases

1. **Bacterial spot** - *Xanthomonas campestris* pv. *vesicatoria*
2. Bacterial wilt - *Ralstonia solanacearum*

Viral diseases

1. **Mosaic** - *Potato virus Y* (PVY) and its strains, *cucumber mosaic virus* (CMV), *tobacco mosaic Virus* (TMV), *tobacco etch virus*, *pepper vein banding virus*, *pepper mottle virus* and *pepper sever mosaic virus*
2. **Leaf curl** - *Tomato leaf curl virus* (TLCV)

Damping off - *Pythium aphanidermatum, P. indicum, , Phytophthora parasitica, Rhizoctonia solani* and *Sclerotium rolfsii.*

Symptoms, pathogen, disease cycle, epidemiology and management are **similar to tomato.**

Die back and fruit rot - *Colletotrichum capsici* (Tel: *Glomerella cingulata)*

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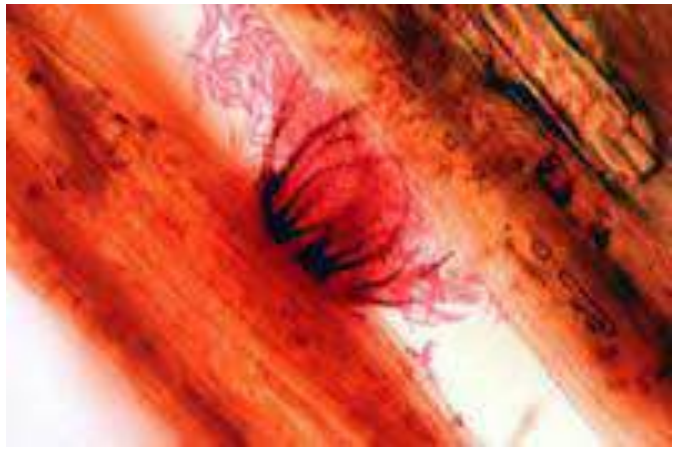

Die back and fruit rot - *Colletotrichum capsici* (Tel: *Glomerella cingulata)*

- Plenty of moisture with moderate temperature favours the disease
- Fruiting coincides with the monsoon rains
- Immediately after rains dew starts which provides sufficient moisture for spore germination and disease assumes serious proportion

Symptoms: The symptoms of the disease appear in two phases *i.e.* die-back; anthracnose and ripe fruit rot.

a) Die-back :

- Symptoms appear **as necrosis of tender twigs from the tip downwards**.
- The entire plant or branch may wither away.
- The twigs become straw coloured in advanced stages of the disease.

Die back and fruit rot - *Colletotrichum capsici* (Tel: *Glomerella cingulata)*

a) **Die-back contd …**

- Large numbers of black dots (acervuli of the fungus) are seen scattered all over the necrotic parts of the plants.
- Terminal and side branches may be killed some times entire plant may be withered.
- Fungus may also attack the stem

Die back and fruit rot - *Colletotrichum capsici* (Tel: *Glomerella cingulata)*

b) Anthracnose and ripe fruit rot:

The disease is noticed on **fully mature green fruits and on red ripe fruits**.

- The symptoms are characterized by the appearance of small, circular, yellowish to pinkish, sunken spots on the skin of the fruits which **spread in the direction of long axis**
- As the fruit matures, these spots enlarge become **brownish to black**, coalesce and form patches on the fruit covering large areas and severely infected fruits look straw coloured and bear numerous dots like acervuli in concentric rings.

Die back and fruit rot - *Colletotrichum capsici* (Tel: *Glomerella cingulata)*

Anthracnose and ripe fruit rot contd….

- Fruits with many lesions may drop prematurely with heavy reduction in yield
- When the fruit is cut open, lower surface of the fruit skin is covered with minute, elevated, spherical, **black stromatic masses of the fungus**
- After invading the skin of the fruit the **fungus enters the seed cavity and infect the seed**
- The seeds produced in such fruits are discoloured and covered with mycelial mat.
- Infected seeds are rusty in colour, when there is **heavy infection in the seed cavity, seeds get shriveled.**

Die back and fruit rot - *Colletotrichum capsici* (Tel: *Glomerella cingulata)*

Disease cycle and epidemiology:

Wide host range: *Alstonia scholaris*, *Tinospora cordifolia*, *Amaranthus spinosus*, *Adathoda vasica*, *Cassia occidentalis*, *Sida cordifolia*, croton, bougainvillae, tomato, turmeric, cotton and bengal gram

- The pathogen survives both in infected plant debris and in the infected seed.
- The fungus can survive in plant debris in the soil at least **for nine months which serve as source of primary infection,** whereas secondary infection takes place through wind borne conidia.
- A temperature of **26° C and presence of free moisture or RH 100 per cent** is optimum for disease development and progress.

Die back and fruit rot - *Colletotrichum capsici* (Tel: *Glomerella cingulata)*

Management:

- Collect and burn the infected plant debris
- One year crop rotation with non solanaceous crops
- Growing resistant varieties viz., **LCA 301, LCA324 and K1**
- Remove and destroy solanaceous weed hosts in and around the field.
- Use seed from healthy fruits. Treat the seeds with captan (0.3%).
- Spray the crop with carbendazim (0.1%) or thiophanate methyl (0.1%) or combination of mancozeb (0.25%) and carbendazim (0.05%) and repeat at 10 to 14 days interval.
- Biocontrol with *Saccharomyces cerevisiae* (seed treatment @5g/lt) followed by ***Bacillus subtilis* (endospores - 10^8) @5ml/lt** at 105 and 120 days after sowing

Powdery mildew - *Leveillula taurica* (anamorph = *Oidiopsis taurica)*

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It is one of the important disease problems of this crop particularly in protected cultivation conditions.

Symptoms:

- On the **under surface of the leaves white to grey-coloured** spots appear while their corresponding upper surface exhibit yellow lesions with brown necrotic centers.
- Affected leaves **curl upwards.**
- Premature senescence of the **leaves results in defoliation**.
- Severe infection may cause **die-back of the twigs or branches and stunting of plants followed by fruit drop**.

Powdery mildew - *Leveillula taurica* (anamorph = *Oidiopsis taurica)*

Disease cycle and epidemiology:

- The pathogen has wide host range including both **cultivated and wild hosts** which ensures the survival of the pathogen from one season to another.
- Temperatures of 25-30° C favour germination of conidia.
- High relative humidity during night than day time and temperatures <30° C are favourable for the disease.
- Low relative humidity *i.e.*, **< 50 per cent favours disease spread**.
- **Free moisture on leaves inhibit spore germination.**

Powdery mildew: *Leveillula taurica* (Lév.) G. Arnaud

Primary spread : pathogen produc[redacted]ospores which infect roots, lower leaves

Symptoms on plants leaves as powdery growth

Secondary spread via conidia to plants

Pathogen overwinter on infected plant debris in the soil as cleistothecia

Powdery mildew - *Leveillula taurica* (anamorph = *Oidiopsis taurica)*

Management:

- Cultural practices like wider plant spacing, sprinkler irrigation and misting in protected structures along with increased air circulation are effective in keeping the disease under check.

- With the initiation of disease spray the crop with systemic fungicides like hexaconazole (0.05%) or carbendazim (0.1%) or difenoconazole (0.03%) and repeat at 10 to 14 days interval.

- **Under protected conditions initiate** the prophylactic sprays before disease initiation with sulphur.

- Application of **monopotassium phosphate and bicarbonates of sodium and potassium** on foliage reduces the disease severity, preventing defoliation and post harvest decay.

Cercospora leaf spot/ frog eye leaf spot - *Cercospora capsici*

In advanced stages, the leaves become yellow and fall off

Cercospora leaf spot initially appears as small circular spots on leaves. these spots enlarge as the disease progresses.

Cercospora leaf spot/ frog eye leaf spot - *Cercospora capsici*

Photo 2. Typical Symptoms of *Cercospora* leaf spot of Chilli fruit

Cercospora leaf spot/ frog eye leaf spot - *Cercospora capsici*

Cercospora leaf spot/ frog eye leaf spot - *Cercospora capsici*

Symptoms:

- On leaves, **symptoms appear as circular spots with brown borders and light coloured, faded or grayish central part, resemble the frogeye** so the disease is also called as frog eye leaf spot.
- When many spots appear on the leaves, such leaves turn yellow and fall down prematurely, **severe defoliation exposes the fruits to sun and results in sunscald**.
- Symptoms also appear on the **stems and petioles as elongated or irregular spots.**
- Symptoms are also observed **on fruits**.

Cercospora leaf spot/ frog eye leaf spot - *Cercospora capsici*

Disease cycle and epidemiology:

- The fungus mainly perennates through the **infected plant debris** in the form of mycelium or stromata.
- The fungus can also overwinter in the form of **infected seed** in case of fruit infection,
- The conidia develop on the **overwintering stromata** which serve as the source of primary infection.
- These conidia are disseminated by wind and rain splashes to the leaf surface where they germinate and cause infection.
- Disease development is favoured by **high humidity (> 95%) and temperature (20- 25°C).**

Cercospora leaf spot frog eye leaf spot - *Cercospora capsici*

Management:

- Collect and burn the infected plant debris.
- **Follow crop rotation and maintain proper drainage** in the field.
- Use **healthy seeds and treat the seeds** with captan (0.3%).
- Spray the crop with carbendazim (0.1%) or thiophanate methyl (0.1%) or combination of mancozeb (0.25%) and carbendazim (0.05%) or difenoconazole (0.03%) and repeat at 10 to 14 days interval.

Alternaria leaf spot/blight - *Alternaria solani*

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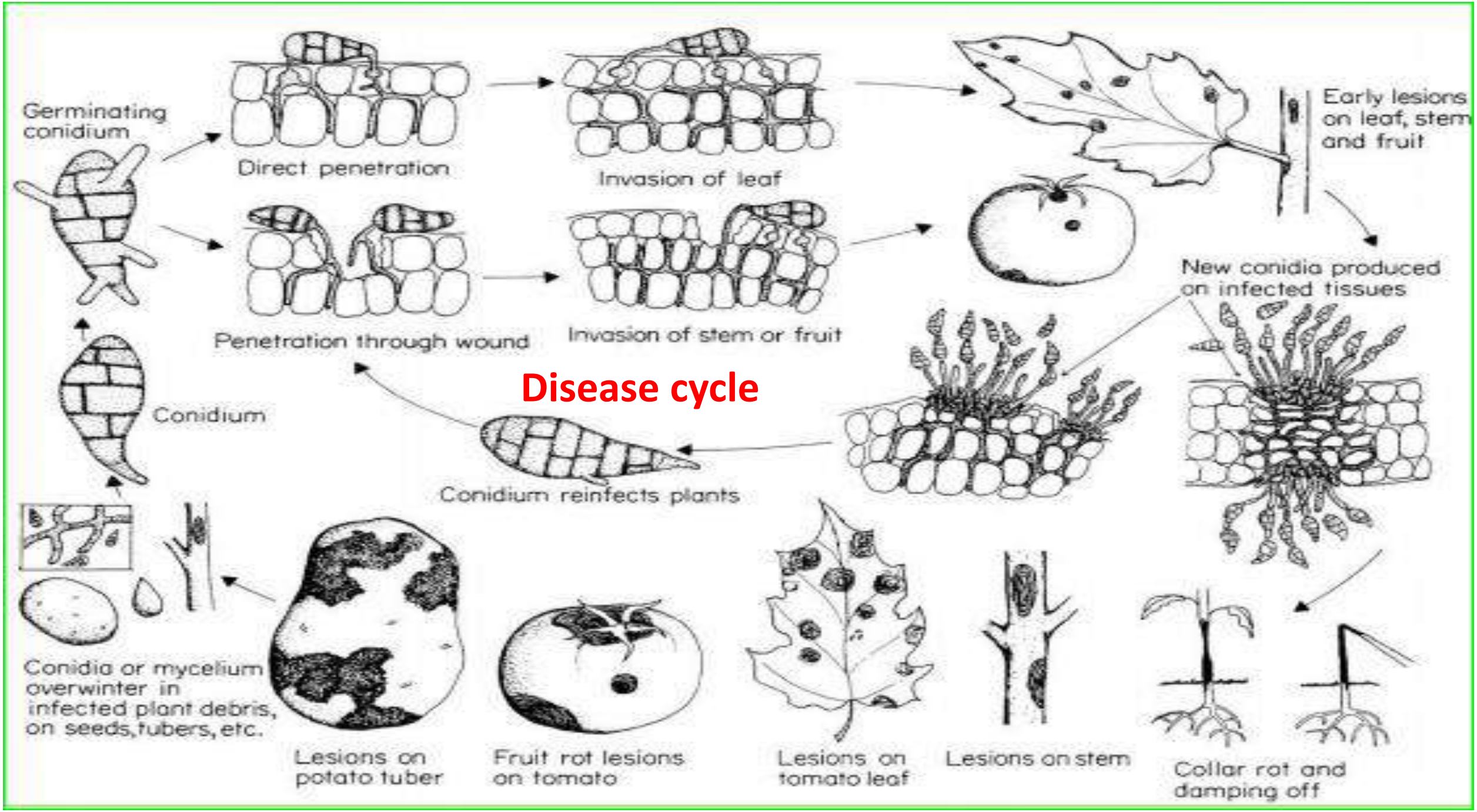
Germinating conidium
Direct penetration
Invasion of leaf
Early lesions on leaf, stem and fruit
Penetration through wound
Invasion of stem or fruit
New conidia produced on infected tissues
Disease cycle
Conidium
Conidium reinfects plants
Conidia or mycelium overwinter in infected plant debris, on seeds, tubers, etc.
Lesions on potato tuber
Fruit rot lesions on tomato
Lesions on tomato leaf
Lesions on stem
Collar rot and damping off

Alternaria leaf spot /blight - *Alternaria solani*

Symptoms:

- **On leaves - Dark, leathery leaf spots** of various sizes on the leaves.
- The infected older leaves die pre-maturely & finally drop.
- **On Fruits** -The spots are covered with black spore masses in concentric circle.
- Management - Refer to Early blight of tomato.

Sclerotial wilt or Stem Rot - *Sclerotium rolfsii*

Symptoms:

- This fungus **infects emerging seedlings below or at the soil level and causes damping-off.**
- Under favorable environmental conditions, *Sclerotium rolfsii* is able to infect any part of the plant.
- The first symptoms of disease in mature plants are **dark-brown lesions on the stem at or just beneath** the soil line.
- First foliar symptoms are **sudden wilting of the plants**.

Sclerotial wilt or Stem Rot - *Sclerotium rolfsii*

Symptoms contd…

- Later, the fungus produces **fan-like web of whitish mycelium** around the rotted stem.
- **Small brown sclerotia form within the mycelial mass**.
- As the disease progresses, infected plants wilt and die.

Management

- Crop rotation with non-host crops like cereals.
- Drenching with chestnut compound (copper sulphate 2 prts + Ammonium carbonate 11 prts) at 3 gm/ltr

Verticillium wilt - *Verticillium dahliae, V. albo-atrum*

Verticillium wilt - *Verticillium dahliae, V. albo-atrum*

Figure 1. Early symptoms of Verticillium wilt in

Begins as pale green blotches between veins and leaf margins. Eventually leaf will wilt, turn brown and die as the disease move up the plant. Often only on one side of the plant

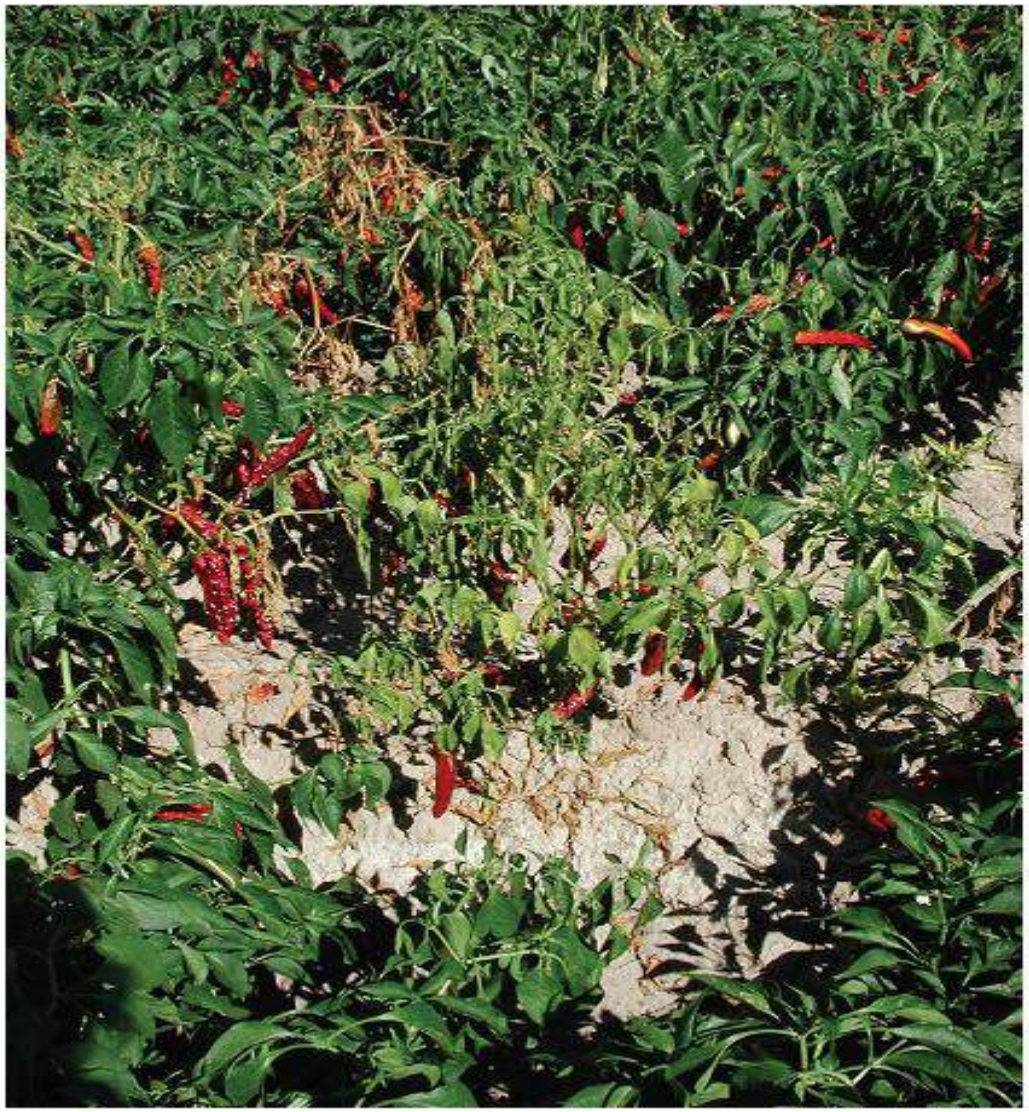

Figure 2. Defoliation caused by *Verticillium*.

Verticillium wilt - *Verticillium dahliae, V. albo-atrum*

Fig. 7. Early stage of Verticillium wilt in a chile plant. Note the slight yellowing of leaves of the affected plant in contrast to the bright green foliage of healthy surrounding plants

Fig. 8. Scattered plants affected by Verticillium wilt in a furrow-irrigated chile field

Stunting of plants may also occur, as shown here with verticillium infected peppers on the right compared to healthy plants on the left

Cutting open the stem reveals brown flecks of discoloured vascular tissue, often in a V-shape

Figure 3. **Vascular discoloration exhibited by chile pepper infected with *Verticillium*.**

Verticillium wilt - *Verticillium dahliae, V. albo-atrum*

Symptoms:

- These soil borne pathogens cause **a wilt by blocking the vascular system** of the plant.
- **Although seedlings can be infected, symptoms usually not observed until plants are older.**
- A characteristic symptom of infection is a **V-shaped lesion that develops on older leaf tips that later expands to cover the leaf**. Infected peppers are stunted and lower leaves are slightly chlorotic.
- In pepper, as the disease progresses stunting and chlorosis become severe with diurnal wilting.

Verticillium wilt - *Verticillium dahliae, V. albo-atrum*

Symptoms contd…

- **Wilting can be asymmetric,** with sections of the plant remaining turgid.
- **Permanent wilt** and plant death follow.
- Dissecting through the crown of affected plants reveals dark-brown vascular discoloration, which can extend into the pith and up into the stem and branches.
- Fruit that form are small and deformed with internal discoloration

Verticillium wilt- *Verticillium dahliae*

Disease Cycle and Epidemiology

- Verticillium wilt is soilborne fungus infects **the roots of host plants**
- Overwinters as a survival structure called a microsclerotium can survive for up to 10 years in the soil without host and withstand drought and cold temperatures.
- Ideal conditions for Verticillium wilt include **cooler weather (75 to 85 ^{o}F) and neutral to alkaline soil (pH of 7 to 9).**
- The **host root exudates trigger germination of microsclerotia** in soil.

Verticillium wilt- *Verticillium dahliae*

Management

- Avoid application of **higher doses of nitrogen**
- Rotation with non host crops
- Mulching with polythene sheet
- Soil fumigation
- Drenching of the soil around the plants with carbendazim 0.1% is recommended

Fusarium wilt -*Fusarium oxysporum* f. sp. *capsici*

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Fusarium wilt -*Fusarium oxysporum* f. sp. *capsici*

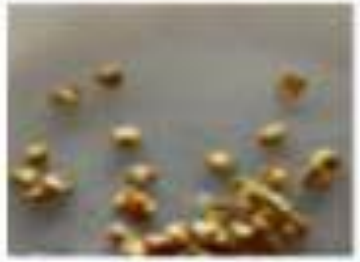
Figure 1. Yellowing of leaves caused by Fusarium wilt.

Fusarium wilt -*Fusarium oxysporum* f. sp. *capsici*

Symptoms:

- Symptoms first appear as a slight yellowing, drooping and withering of foliage.
- As wilting progresses, leaves may turn dull-green to brown and remain attached to the plant.
- When the stem and roots are cut diagonally, reddish-brown streaks are visible in the vascular tissues

FUSARIAL WILT OF CHILLI



Primary spread through seed, soil, water, seedling, workers etc.

Symptoms on plants stems, leaves or fruits

Secondary spread through irrigation water , windfrom infected plantdebris

Pathogen survives on seeds or plant debris and other hostplants

Fusarium wilt -*Fusarium oxysporum* f. sp. *capsici*

Favourable conditions

- *Fusarium* wilt is a warm weather soil borne fungal disease that does optimally at **temperatures of about 28°C (25-30°C).**
- **Hot dry weather** favour wilt development
- More severe in **acidic soils (5.6 to 6.5 pH) than in alkaline ones**
- Presence of **root knot nematodes and monoculture** enhance the disease further
- Favoured by **ammonical form** of nitrogen

Fusarium wilt -*Fusarium oxysporum* f. sp. *capsici*

Management

- Avoid application of **higher doses of nitrogen**
- Rotation with **non host crops**
- Mulching with polythene sheet
- Soil fumigation
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